BIKASH SAGAR KOIRI

Durgapur, West Bengal   +91-6297046553   sagarbikash97@gmail.com
bikashsk26 | Bikash07-git

Professional Summary

M.Tech Computer Science student with expertise in Data Analytics, Machine Learning, and Business Intelligence. Proficient in Python, SQL, Power BI, Excel, and statistical analysis, with hands-on experience in EDA, dashboard development, predictive modeling, and transforming complex data into actionable business insights. Published IEEE conference researcher seeking opportunities in Data Analytics, Technology Risk, and Business Technology Solutions.

Technical Skills

Programming: Python, SQL, C++
Data Analysis: EDA, Data Cleaning, Statistical Analysis, Feature Engineering, KPI Reporting
Visualization: Power BI, Excel, DAX, Power Query, Dashboard Development
Machine Learning: Scikit-learn, Classification, Regression, Cross-Validation, Hyperparameter Tuning, SHAP
Databases: MySQL, RDBMS, SQL Joins, CTEs, Window Functions
Libraries & Tools: Pandas, NumPy, Matplotlib, Plotly, Git, GitHub, Jupyter Notebook, Google Colab, VS Code

Projects

Customer Churn Analysis & Prediction |Github — *May 2026 – Jun 2026*

Python, Pandas, NumPy, Power BI, Matplotlib, Seaborn, Data Analytics

- Analyzed **7,032 customer records** across **21 features** to identify factors driving a **26.58% churn rate**.
- Performed **data cleaning, preprocessing, and EDA** to uncover customer behavior, service usage, and billing patterns.
- Built a **4-page Power BI dashboard** with **20 visualizations** and **4 KPI metrics** for churn analysis and executive reporting.
- Identified high-risk customer segments and generated **15+ actionable insights** to support retention strategies and business decisions.

Sales Intelligence Dashboard with Predictive Analytics |Github — *Nov 2025 – Dec 2025*

Python, SQL, Pandas, NumPy, Plotly, Power BI, Git

- Analyzed **1,000+ retail transactions** using **Python, SQL, and Pandas**.
- Developed **Power BI dashboards** and predictive models for sales trend analysis.
- Identified key revenue drivers, with top categories contributing **30%+ of total sales**.

Chronic Kidney Disease Prediction using Machine Learning |Github — *Sep 2025 – Oct 2025*

Python, Scikit-learn, SHAP, NumPy, Pandas

- Developed an end-to-end **ML pipeline** on a **425-record clinical dataset**.
- Increased prediction accuracy from **87% to 93%** through **hyperparameter optimization**.
- Implemented **SHAP** for model explainability and transparent predictions.

Publications & Research

Pathological Voice Disorder Detection – IEEE ETAACT'26 | IEEE Xplore — *Jan 2026 – Apr 2026*

SVM, MFCC, LPCC, Formants, Audio Signal Processing, Feature Extraction

- Developed a **Progressive Acoustic Stacking (PAS)** framework for pathological voice disorder classification using advanced audio signal processing techniques.
- Extracted and progressively integrated **MFCC, LPCC, Formant, and voice quality features** from a **3,677-sample voice dataset**.
- Built and optimized an **SVM-based machine learning model**, achieving **87.32% speaker-level accuracy and an 81.58% F1-score**.
- **Outperformed baseline models by 4.17%**, with the research paper accepted at **IEEE ETAACT'26 for publication in IEEE Xplore**.

Education

Birla Institute of Technology, Mesra — *2024 – 2026*
Master of Technology — Computer Science and Engineering | ***CGPA: 8.39*** — Jharkhand, India

National Institute of Science and Technology, Berhampur — *2020 – 2024*
Bachelor of Technology — Computer Science and Engineering | ***CGPA: 8.15*** — Odisha, India

Certifications & Achievements

- **Published IEEE research paper** in Machine Learning | Certificate — 2026
- **Data Analyst Certification** — oneroadmap | Certificate — 2026
- **EY Technology Risk Virtual Job Simulation** | Certificate — 2026
- **SQL** — HackerRank | Certificate — 2026